

DETAILED NOTES: DIFFERENTIATION

Mathematics — Differential Calculus

1. Introduction to Differentiation

Differentiation is one of the most important topics in Calculus. It is used to study rate of change, slopes, maxima/minima, motion, optimization, approximation, and many applications in engineering, physics, economics, and computer science.

If $y = f(x)$, the derivative of y with respect to x measures how rapidly y changes when x changes.

Derivative notations: $\frac{dy}{dx}$, $f'(x)$, y' , and $D_x(y)$.

Example: If $y = x^2$, then $\frac{dy}{dx} = 2x$.

2. Geometrical Meaning of Derivative

The derivative represents the slope of the tangent to a curve at a particular point.

Slope of tangent = $\frac{dy}{dx}$

If $\frac{dy}{dx} > 0$, the function is increasing. If $\frac{dy}{dx} < 0$, the function is decreasing. If $\frac{dy}{dx} = 0$, the tangent is horizontal.

3. Derivative from First Principles

For $y = f(x)$, the derivative is defined using a limit:

$$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$$

This is called the derivative from first principles.

Example: Derivative of x^2 from First Principles

Let $f(x) = x^2$.

$$f'(x) = \lim_{h \rightarrow 0} [(x+h)^2 - x^2] / h$$

$$= \lim_{h \rightarrow 0} [x^2 + 2xh + h^2 - x^2] / h$$

$$= \lim_{h \rightarrow 0} (2x + h)$$

Therefore, $f'(x) = 2x$.

4. Basic Rules of Differentiation

4.1 Constant Rule

If c is a constant: $\frac{d(c)}{dx} = 0$. Example: $\frac{d(15)}{dx} = 0$.

4.2 Power Rule

If $y = x^n$, then $\frac{dy}{dx} = n x^{n-1}$.

Examples: $\frac{d(x^5)}{dx} = 5x^4$; $\frac{d(x^{10})}{dx} = 10x^9$; $\frac{d(\sqrt{x})}{dx} = 1/(2\sqrt{x})$.

4.3 Constant Multiple Rule

If $y = c f(x)$, then $\frac{dy}{dx} = c f'(x)$. Example: $\frac{d(5x^4)}{dx} = 20x^3$.

4.4 Sum and Difference Rule

$$\frac{d[f(x) \pm g(x)]}{dx} = f'(x) \pm g'(x).$$

Example: If $y = x^4 + 3x^2 - 7x + 10$, then $y' = 4x^3 + 6x - 7$.

5. Product Rule

When two functions are multiplied, $y = u(x)v(x)$:

$$(uv)' = uv' + vu'$$

Example: $y = x^2 \sin x$. Here $u = x^2$, $v = \sin x$, $u' = 2x$, $v' = \cos x$.

Therefore, $y' = x^2 \cos x + 2x \sin x$.

6. Quotient Rule

If $y = u/v$, then:

$$(u/v)' = (v u' - u v') / v^2$$

Memory aid: Low $\times$ derivative of High $-$ High $\times$ derivative of Low, divided by Low².

Example: $y = x^2/(x+1)$. Then $y' = [(x+1)(2x) - x^2]/(x+1)^2 = (x^2+2x)/(x+1)^2$.

7. Chain Rule

The chain rule is used when one function is inside another. If $y = f(g(x))$, then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

In simple words: differentiate the outer function, keep the inner function unchanged, then multiply by the derivative of the inner function.

Example: $y = (3x^2+2)^5$. Therefore $y' = 5(3x^2+2)^4(6x) = 30x(3x^2+2)^4$.

8. Differentiation of Algebraic Functions

For polynomial functions, differentiate each term using the power rule.

Example: $y = 3x^4 - 4x^3 + 7x - 9$ gives $y' = 12x^3 - 12x^2 + 7$.

For fractional powers, $y = x^{(m/n)}$, so $y' = (m/n)x^{(m/n)-1}$. Example: $\frac{d(x^{3/2})}{dx} = (3/2)x^{1/2}$.

9. Differentiation of Exponential Functions

$$\frac{d(e^x)}{dx} = e^x$$

For $y = a^x$: $\frac{dy}{dx} = a^x \ln(a)$.

Example: $\frac{d(2^x)}{dx} = 2^x \ln 2$.

10. Differentiation of Logarithmic Functions

$$\frac{d(\ln x)}{dx} = 1/x$$

More generally: $\frac{d[\ln f(x)]}{dx} = f'(x)/f(x)$.

Example: $y = \ln(x^2+1)$, so $y' = 2x/(x^2+1)$.

For logarithm base a : $d(\log_a x)/dx = 1/(x \ln a)$.

11. Differentiation of Trigonometric Functions

Function	Derivative
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$

12. Trigonometric Functions with Chain Rule

Example: $y = \sin(3x)$. Then $y' = \cos(3x) \cdot 3 = 3 \cos 3x$.

Example: $y = \cos(x^2)$. Then $y' = -\sin(x^2) \cdot 2x = -2x \sin(x^2)$.

13. Inverse Trigonometric Functions

Function	Derivative
$\sin^{-1} x$	$1/\sqrt{1-x^2}$
$\cos^{-1} x$	$-1/\sqrt{1-x^2}$
$\tan^{-1} x$	$1/(1+x^2)$
$\cot^{-1} x$	$-1/(1+x^2)$
$\sec^{-1} x$	$1/(x \sqrt{x^2-1})$
$\csc^{-1} x$	$-1/(x \sqrt{x^2-1})$

14. Differentiation of Composite Functions

A composite function has the form $y = f(g(x))$. Apply the chain rule.

Example: $y = \sqrt{5x+1} = (5x+1)^{1/2}$.

$y' = (1/2)(5x+1)^{-1/2}(5) = 5/[2\sqrt{5x+1}]$.

15. Implicit Differentiation

Sometimes y is not explicitly given as a function of x . For example: $x^2 + y^2 = 25$.

Differentiate both sides with respect to x : $2x + 2y(dy/dx) = 0$.

Therefore, $dy/dx = -x/y$.

Important: whenever differentiating a term containing y , remember that y is a function of x . For example, $d(y^2)/dx = 2y(dy/dx)$.

16. Parametric Differentiation

If $x = f(t)$ and $y = g(t)$, then:

$$dy/dx = (dy/dt)/(dx/dt)$$

Example: $x = t^2$, $y = t^3$. Then $dx/dt = 2t$ and $dy/dt = 3t^2$. Hence $dy/dx = 3t/2$.

17. Logarithmic Differentiation

Logarithmic differentiation is useful when the variable occurs in both the base and exponent or when products/powers are complicated.

Example: $y = x^x$

Take logarithms: $\ln y = x \ln x$.

Differentiate: $(1/y)(dy/dx) = \ln x + 1$.

Since $y = x^x$, $dy/dx = x^x(\ln x + 1)$.

18. Higher-Order Derivatives

The derivative of a derivative is called a higher-order derivative.

First derivative: dy/dx . Second derivative: d^2y/dx^2 . Third derivative: d^3y/dx^3 .

Example: $y = x^4 \rightarrow y' = 4x^3 \rightarrow y'' = 12x^2 \rightarrow y''' = 24x \rightarrow y^{(4)} = 24 \rightarrow y^{(5)} = 0$.

19. Increasing and Decreasing Functions

If $f'(x) > 0$, the function is increasing. If $f'(x) < 0$, the function is decreasing. If $f'(x) = 0$, the point may be a stationary point.

20. Critical Points

A critical point occurs where $f'(x) = 0$ or where $f'(x)$ does not exist, provided x is in the domain of the function.

Example: $f(x) = x^2 - 4x + 3$. Then $f'(x) = 2x - 4$. Setting $f'(x) = 0$ gives $x = 2$.

21. Maxima and Minima

To find local maxima/minima: (1) find $f'(x)$, (2) solve $f'(x) = 0$, (3) find $f''(x)$, and (4) apply the second derivative test.

If $f''(x) > 0$ at a critical point, it is a local minimum. If $f''(x) < 0$, it is a local maximum. If $f''(x) = 0$, the test is inconclusive.

Example

For $f(x) = x^2 - 4x + 5$, $f'(x) = 2x - 4$, giving $x = 2$. Since $f''(x) = 2 > 0$, there is a minimum at $x = 2$. The minimum value is $f(2) = 1$.

22. Tangent and Normal

At (x_1, y_1) , the slope of tangent is $m = (dy/dx)$ at $x=x_1$.

Equation of tangent: $y-y_1 = m(x-x_1)$

Equation of normal: $y-y_1 = -(1/m)(x-x_1)$

23. Rate of Change

If distance s is a function of time t , then velocity is $v = ds/dt$ and acceleration is $a = dv/dt = d^2s/dt^2$.

Distance → **derivative with respect to time** → **Velocity** → **derivative with respect to time** → **Acceleration**

24. Differentiation Formula Sheet

Rule	Formula
Constant	$(c)' = 0$
Power	$(x^n)' = nx^{n-1}$
Constant multiple	$(cf)' = cf'$
Sum	$(f+g)' = f' + g'$
Difference	$(f-g)' = f' - g'$
Product	$(fg)' = fg' + gf'$
Quotient	$(f/g)' = (gf' - fg')/g^2$
Chain	$[f(g(x))]' = f'(g(x))g'(x)$
Exponential	$(e^x)' = e^x$
Logarithm	$(\ln x)' = 1/x$

25. Solved Mixed Examples

Example 1: $d/dx(5x^4 - 3x^2 + 7) = 20x^3 - 6x$.

Example 2: $d/dx[(x^2+1)^3] = 6x(x^2+1)^2$.

Example 3: $d/dx(x^2e^x) = e^x(x^2+2x)$.

Example 4: $d/dx(\sin x/x) = (x \cos x - \sin x)/x^2$.

Example 5: $d/dx[\ln(\sin x)] = \cot x$.

26. Common Mistakes to Avoid

1. Forgetting the coefficient in the power rule: $d(x^5)/dx$ is $5x^4$, not x^4 .
2. Forgetting the chain-rule factor. For $(3x+1)^5$, the derivative is $15(3x+1)^4$.
3. Using $(uv)' = u'v'$. The correct product rule is $(uv)' = uv' + vu'$.
4. Forgetting the negative sign in $(\cos x)' = -\sin x$.
5. Confusing $\sin^{-1}x$ with $1/\sin x$. In calculus, $\sin^{-1}x$ generally means $\arcsin x$.

27. Problem-Solving Strategy

Step 1: Identify the function type — polynomial, product, quotient, composite, exponential, logarithmic, trigonometric, implicit, or parametric.

Step 2: Select the appropriate differentiation rule.

Step 3: Differentiate carefully.

Step 4: Simplify.

Step 5: Substitute values if required.

28. Important Exam Questions

1. Define differentiation and explain its geometrical meaning.
2. Derive the derivative of x^2 from first principles.
3. State and apply the power, product, quotient, and chain rules.
4. Differentiate algebraic, exponential, logarithmic, and trigonometric functions.
5. Explain implicit and parametric differentiation with examples.
6. Explain logarithmic differentiation and differentiate x^x .
7. Find higher-order derivatives.
8. Find equations of tangent and normal.
9. Find critical points and determine maxima/minima.
10. Determine intervals where a function is increasing or decreasing.
11. Solve rate-of-change and optimization problems.

29. Quick Revision Cheat Sheet

Power: $d(x^n)/dx = nx^{n-1}$

Product: $d(uv)/dx = uv' + vu'$

Quotient: $d(u/v)/dx = (vu' - uv')/v^2$

Chain: $d[f(g(x))]/dx = f'(g(x))g'(x)$

Exponential: $d(e^x)/dx = e^x$; $d(a^x)/dx = a^x \ln a$

Logarithmic: $d(\ln x)/dx = 1/x$

Trig: $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$; $(\tan x)' = \sec^2 x$

Inverse trig: $(\sin^{-1} x)' = 1/\sqrt{1-x^2}$; $(\tan^{-1} x)' = 1/(1+x^2)$

Applications: $f'(x)=0$ gives possible critical points; $f''(x)>0$ indicates local minimum; $f''(x)<0$ indicates local maximum.